MMML Accuracy-Robustness Tradeoff

Terminologies

- **Adversarial examples** synthesize small, imperceptible perturbations of the input data and cause the model to make highly-confident but erroneous predictions [1] and usually come in the form of **restricted attacks** (ϵ-bounded perturbations) or **unrestricted attacks** (adversarial rotations, translations, deformations) [5].
- **Adversarial training** repeatedly finds the worst perturbations and updates the model to reduce the loss on perturbed inputs, which requires more training time/data [1].
- **Transferability** is the phenomenon that one set of adversarial examples is likely to fool other classifiers learned from samples of the same distribution [1].
- **Inductive bias** is the set of extra assumptions that the learning algorithm makes about the target function that enables it to generalize beyond the training data [3].
- **Canonical Polyadic (CP)-decomposition**: for an order-M tensor $\mathcal{X} \in \mathbb{R}^{d_1 \times \cdots \times d_M}$, there exists an exact decomposition into vectors $\mathbf{w}$:

  $$\mathcal{X} = \sum_{i=1}^{r} \bigotimes_{m=1}^{M} \mathbf{w}_m^i$$

  The minimal r for exact decomposition is the **rank** of the tensor, which measures how many vectors are required to reconstruct the tensor. The vectors $\{\{\mathbf{w}_m^i\}_{m=1}^M\}_{i=1}^r$ are called the **rank r decomposition factors** of $\mathcal{X}$ [4].
- **Nuclear norm** of an order-M tensor is defined as [4]

  $$\|\mathcal{X}\|_* = \inf\left\{\sum_{i=1}^{r} |\lambda_i| : \mathcal{X} = \sum_{i=1}^{r} \lambda_i \left(\bigotimes_{m=1}^{M} \mathbf{w}_m^i\right), \|\mathbf{w}_m^i\| = 1, r \in \mathbb{N}\right\}$$

  The tensor **Frobenius norm scaled by the tensor dimensions** upper bounds the nuclear norm and can be efficiently computed, defined as

  $$\|\mathcal{M}\|_* \leq \sqrt{\frac{\prod_{i=1}^{M} d_i}{\max\{d_1, \ldots, d_M\}}} \|\mathcal{M}\|_F$$

- Let X be a real topological vector space and X^* be the dual space to X, for a function $f : X \to \mathbb{R} \cup \{-\infty, +\infty\}$, the **convex conjugate** is defined as $f^*(\mathbf{v}) = \sup_{\mathbf{u}}\{\mathbf{u}^T\mathbf{v} - f(\mathbf{u}) \mid \mathbf{u} \in X\}$ [5].

- For a **surrpogate loss** ϕ and $\eta \in [0, 1]$, define the **conditional** ϕ**-risk** by $H(\eta) = \inf_{\alpha\in\mathbb{R}} C_\eta(\alpha) = \inf_{\alpha\in\mathbb{R}}(\eta\phi(\alpha) + (1-\eta)\phi(-\alpha))$ and $H^-(\eta) = \inf_{\alpha(2\eta-1)\leq 0} C_\eta(\alpha)$.

  The loss is **classification-calibrated** if the optimal classifier under ϕ is optiml under the Bayes decision rule, or imposing the constraint that α has an inconsistent sign with the Bayes decision rule $\mathrm{sgn}(2\eta - 1)$ leads to strictly larger ϕ-risk. Then for any $\eta \neq \frac{1}{2}, H^-(\eta) > H(\eta)$ [5].

- **Projected gradient descent** minimizes a function subject to a constraint by moving in the direction of negative gradient and projecting onto the feasible set at each step [5].

- **Fast Gradient Sign Method (FGSM)** is one of the baseline attacks to deep neural networks, which computes an adversarial example as $\mathbf{x}' = \mathbf{x} + \epsilon\mathrm{sgn}(\nabla_{\mathbf{x}}\phi(f(\mathbf{x})y))$ where $f : \mathcal{X} \to \mathbb{R}$ is the **score function** that maps instance to its confidence value of being positive and $\phi(\cdot)$ is a surrogate of 0-1 loss.

  The multi-step variant **FGSM^k / PGD attack** applies projected gradient descent k times:

$$\mathbf{x}'_{t+1} = \Pi_{\mathbb{B}(\mathbf{x},\epsilon)}(\mathbf{x}'_t + \epsilon\mathrm{sgn}(\nabla_{\mathbf{x}}\phi(f(\mathbf{x}'_t)y)))$$

  where $\Pi_{\mathbb{B}(\mathbf{x},\epsilon)}$ is the projection operator onto $\mathbb{B}(\mathbf{x}, \epsilon)$. Both FGSM and FGSM^k approximately solve $\max_{\mathbf{x}'\in\mathbb{B}(\mathbf{x},\epsilon)} \phi(f(\mathbf{x}')y)$.

- **Robust optimization based defense** trains a network by fitting the parameters to the adversarial examples

$$\min_f \mathbb{E}\left\{\max_{\mathbf{X}'\in\mathbb{B}(\mathbf{X},\epsilon)} \phi(f(\mathbf{X}')Y)\right\}$$

  but lack theoretical guarantees and don't consider the accuracy-robustness tradeoff.

Week 1

1 Robustness May Be at Odds with Accuracy

Challenge/Opportunity

- Many proposed robust training methods are ineffective.
- Computational complexity and training data may not be the only costs of adversarial robustness, and it remains a question if robust model is always preferable than a standard model.
- Empirical evidence shows that there exists a trade-off between standard and adversarially robust accuracy.

SOA Knowledge/Solution

- It is already known that robustness comes at a cost of computational complexity and more training data.
- The most successful robust training approach so far is adversarial training.
- Adversarial perturbation is commonly mistaken as a form of data augmentation that benefits standard accuracy.
- Previous works prove upper bounds on the robustness of classifiers and show regularized gradients are more human-aligned and give better adversarial examples.

Approach

- The trade-off between standard accuracy and robustness is demonstrated in a simple binary classification task.

Results

- The paper concludes that the trade-off is caused by the fundamental difference in the feature representations learned by standard and robust models.
- Robustness is unlikely resulted from standard training, so robust training methods are important.
- Robust models learn more human-aligned feature representations.
- Robust models yield clean feature interpolations similar to those obtained from generative models.

Thoughts

- This trade-off between standard and adversarially robust accuracy is significant since it contradicts a misconception that sufficient amount of training data will build a robust model. There are two aspects in terms of training a model: feature selection and feature learning. The paper shows that robustness is about the former while the amount of training data affects the latter.
- Robust training methods are more connected to generative networks like GAN since both focus on selecting and twisting the most correlated feature to confuse the model/discriminator.
- Humans can have lower accuracy in certain benchmarks than ML models potentially because ML models rely on brittle features that humans are naturally invariant to. So is higher standard accuracy really an indicator for better model?
- Robustness may be a better indicator because of cross-model adversarial transferability.
- Adversarial training may be sufficient for attaining more human-aligned gradients.

2 Found in Translation: Learning Robust Joint Representations by Cyclic Translations Between Modalities

Challenge/Opportunity

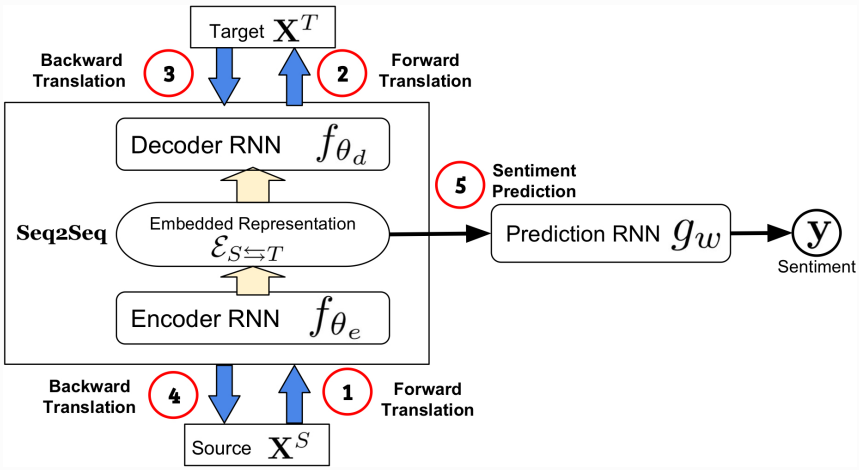
- For multimodal sentiment analysis, existing works require all modalities to learn a joint representation, which is sensitive to noisy or missing modalities during testing.

SOA Knowledge/Solution

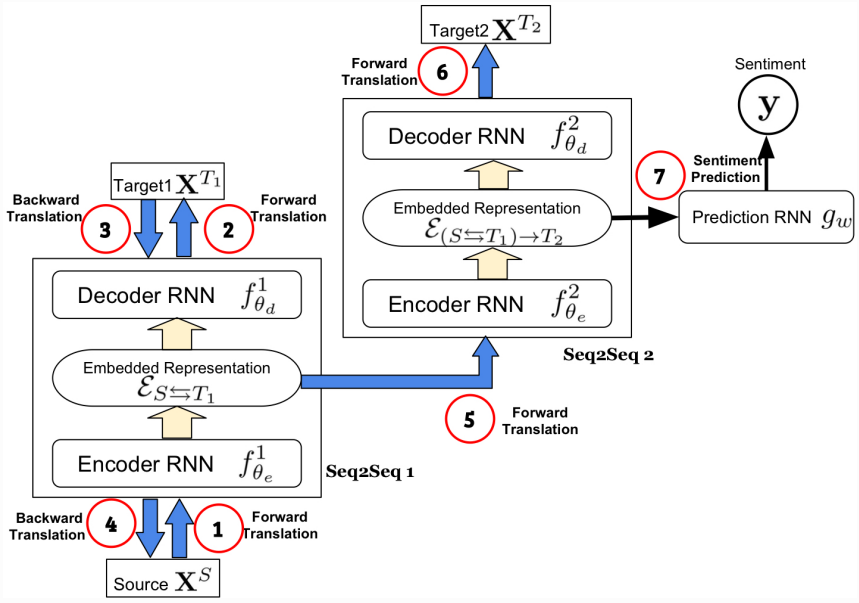
- Earlier works include fusion approaches, multistage approach to learn hierarchical representations, attention and memory mechanism, and generative methods.
- To infer missing modalities, an approach is to model the probabilistic inter-modal relationships. This is achieved by sampling from the conditional distributions over each modality.
- Training with modalities dropped at random improves robustness of joint representations.

Approach (MCTN)

- **Dataset**: CMU-MOSI, ICT-MMMO, and Youtube with the same speaker not appearing in both training and testing.
- **Feature extraction**: GloVe word embeddings (language), Facet (visual), and COVAREP (acoustic), P2FA (alignment)
- **Multimodal cyclic translations**:
  - **Forward translations** (source $\rightarrow$ target): given source modality S and target T,
    - Compute joint embedding (Seq2Seq) $\mathcal{E}_{S\rightarrow T} = f_{\theta_e}(\mathbf{X}^S)$
    - Transform into target modality (beam search approach) $\mathbf{X}^T = f_{\theta_d}(\mathcal{E}_{S\rightarrow T})$
    - Loss: $\mathcal{L}_t = \mathbb{E}[\ell_{\mathbf{X}^T}(\hat{\mathbf{X}}^T, \mathbf{X}^T)]$
  - **Backward translations** (predicted target $\rightarrow$ source):
    - Compute joint embedding $\mathcal{E}_{T\rightarrow S} = f_{\theta_e}(\hat{\mathbf{X}}^T)$
    - Transform into target modality $\hat{\mathbf{X}}^S = f_{\theta_d}(\mathcal{E}_{T\rightarrow S})$
    - Loss: $\mathcal{L}_c = \mathbb{E}[\ell_{\mathbf{X}^S}(\hat{\mathbf{X}}^S, \mathbf{X}^S)]$
  - **Prediction**:
    - Loss: $\mathcal{L}_p = \mathbb{E}[\ell_{\mathbf{y}}(\hat{\mathbf{y}}, \mathbf{y})]$
  - **Coupled translation-prediction objective function**: $\mathcal{L} = \lambda_t \mathcal{L}_t + \lambda_c \mathcal{L}_c + \mathcal{L}_p$



- **Hierarchical MCTN** performs two levels of modality translations and doesn't use a cyclic translation loss except the first level of translation since the ground truth embedding is unknown.



Results

- MCTN chieves new SOA results on the tested datasets.
- MCTN only uses source modality for prediction and thus ensures robustness to noisy/missing modalities.
- Adding more modalities improves learning more discriminative joint representations.
- Cyclic translations add regularization while ensuring the representation retain maximal information from all modalities.
- Language is the most discriminative modality in the joint representation.
- Representation learning is easier when task is broken down recursively.

Thoughts

- Other than selecting the most significant features in learning, another aspect of model robustness is to handle imperfect training/testing datasets.

3 Understanding and Mitigating the Tradeoff Between Robustness and Accuracy

Challenge/Opportunity

- Adversarial training with perturbed training set improves robust error but often increases the standard error.

SOA Knowledge/Solution

- Previous explanations are based on an inherent accuracy-robustness tradeoff that persists even in the infinite data limit and show that standard and robust error are fundamentally at odds.
- **Robust self-training (RST)** is introduced as a robust variant of self-training that overcomes the sample complexity barrier of learning a model with low robust error by leveraging extra unlabeled data.
- **Interpolated Adversarial Training (IAT)** that considers a different training algorithm based on Mixup and **Neural Architecture Search (NAS)** that uses RL to search for more robust architectures were proposed to mitigate the tradeoff between standard and robust error empirically.

Approach

- Consider a linear model where the true linear function has zero standard and robust error with adversarial training augmenting with extra data where the perturbations are consistent: $P_y(\cdot \mid x_{\text{ext}}) = P_y(\cdot \mid x)$.
- Compare the standard errors of the standard estimator and the augmented estimator in noiseless linear regression and characterize the general conditions for standard error not increasing in fitting augmented data.
- Examine the general RST estimator in the linear regression setting, study the effect of the labeled training set sizes on RST improving standard and robust error over vanilla adversarial training, and evaluate RST empirically.

  RST formulation:

  1. Perform standard training on labeled data $\{(x_i, y_i\}_{i=1}^n$ to obtain a standard estimator $\hat{\theta}_{\text{std}} = \arg\min_\theta \sum_{i=1}^n \ell(f_\theta(x_i), y_i)$
  2. Perform robust training on both the labeled and unlabeled inputs $\{\hat{x}_i\}_{i=1}^m$ with pseudo-labels $\hat{y}_i = f_{\hat{\theta}_{\text{std}}}(\hat{x}_i)$ and combined loss to obtain

$\hat{\theta}_{\text{rst}} = \arg\min_\theta(\alpha\hat{L}_{\text{std-lab}}(\theta) + \beta\hat{L}_{\text{rob-lab}}(\theta) + \gamma\hat{L}_{\text{std-unlab}}(\theta;\hat{\theta}_{\text{std}})) + \lambda\hat{L}_{\text{rob-unlab}}(\theta;\hat{\theta}_{\text{std}}))$ for fixed $\alpha, \beta, \gamma, \lambda \geq 0$

Results

- The mismatch between the inductive bias of models and the underlying distribution of the inputs causes the standard error to increase even when the augmented data is perfectly labeled.
    - An augmentation that encourages the fitting local components can potentially increase the error along other global components with larger eigenvalue in Σ results in larger standard error of $\hat{\theta}_{\text{aug}}$.
    - For a large increase in standard error upon augmentation, the true parameter needs to be sufficiently more complex than $\hat{\theta}_{\text{std}}$.
    - Conditions for the standard error not increasing when fitting augmented data:
        1. The population covariance $\Sigma = \mathbb{E}_{P_x}[xx^T]$ is identity.
        2. The augmented data $[X_{\text{std}}; X_{\text{ext}}]$ spans the entire space.
        3. The extra data $x_{\text{ext}} \in \mathbb{R}^d$ is a single point such that it's an eigenvector of Σ.
- RST upperbounds the $L_{\text{std}}(\hat{\theta}_{\text{rst}})$ by $L_{\text{std}}(\hat{\theta}_{\text{std}})$ and makes $L_{\text{rob}}(\hat{\theta}_{\text{rst}}) = L_{\text{std}}(\hat{\theta}_{\text{rst}})$ by incorporating a Σ-induced regularizer while satisfying the robust losses constraints.
- Empirical evidence shows that RST improves both the standard and robust error and mitigate the tradeoff between accuracy and robustness.

Thoughts

- This paper complements [1] by explaining the tradeoff in another theoretical setting and showing the tradeoff doesn't exist in infinite data.

  What is the significance of the noiseless linear regression setting?
- Another reason for the accuracy-robustness tradeoff is **overparametrization.**
- A natural strategy to mitigate this tradeoff is to regularize the augmented estimator to be closer to the standard estimator (which incorporates information about Σ).

- RST captures the global structure and obtains low standard error by fitting extra inputs while enforcing invariance on local transformations on both labeled and unlabeled inputs for obtaining low robust error by capturing the local structure across the domain.

4 Learning Representations from Imperfect Time Series Data via Tensor Rank Regularization

Challenge/Opportunity

- Natural multimodal data are often imperfect (incomplete due to mismatched modalities or sensor failure, or corrupted with random/structured noise), which breaks the temporal correlations and gives tensor representations of higher rank.
- Computing the rank of a tensor or its nuclear norm is NP-hard for tensors of order ≥ 3.

SOA Knowledge/Solution

- Recent research investigated the use of tensors for representation learning: given representations $\mathbf{h}_1, \ldots, \mathbf{h}_M$ from M modalities, the order-M outer product tensor $\mathcal{T} = \mathbf{h}_1 \otimes \mathbf{h}_2 \otimes \cdots \otimes \mathbf{h}_M$ represents all possible interactions between the modality dimensions.
- Generative approaches as neural models (cascaded residual autoencoders), deep adversarial learning, and translation-based learning have been proposed to account for imperfect data, but these methods often require prior knowledge of which entries or modalities are imperfect.

Approach (T2FN)

- Given time series data from the language, visual, and acoustic modalities, **Temporal Tensor Fusion Network (T2FN)** uses LSTM networks to encode the temporal information from each modality, which results in a sequence of hidden representations.
- Form tensors via outer products of the individual representation through time and append a 1 to capture unimodal, bimodal, and trimodal interactions

$$\mathcal{M} = \sum_{t=1}^{T} \begin{bmatrix} \mathbf{h}_\ell^t \\ 1 \end{bmatrix} \otimes \begin{bmatrix} \mathbf{h}_v^t \\ 1 \end{bmatrix} \otimes \begin{bmatrix} \mathbf{h}_a^t \\ 1 \end{bmatrix}$$

- Since the rank of a tensor or its nuclear norm is NP-hard to compute, minimize the tensor Frobenius norm scaled by the tensor dimensions as the upper bound.
- Study the effect of imperfect data on the rank of tensor $\mathcal{M}$ with clear, random drop, and structured drop noises by performing CP decomposition on the tensor representations under different rank settings r and measure the reconstruction error

$$\epsilon = \min_{\mathbf{w}_m^i} \left\| \left(\sum_{i=1}^{r} \bigotimes_{m=1}^{M} \mathbf{w}_m^i \right) - \mathcal{X} \right\|_F$$

  Given the true rank r^*, ϵ will be high at ranks $r < r^*$ and approximately zero at ranks $r \geq r^*$.

Results

- The rank of $\mathcal{M}$ is maximized when data entries are sampled from i.i.d. noise; clean real-world data is often generated from lower dimensional latent structures and multimodal time series data exhibits correlations across time and modalities, giving a low-rank tensor representations; $r_{\text{clean}} < r_{\text{imperfect}} < r_{\text{noisy}}$.
- Imperfection increases tensor rank.
- T2FN as a regularization method based on **tensor rank minimization** learns robust representations when imperfections that increase tensor rank are introduced.

Thoughts

- Tensor rank regularization is a method for learning robust model in the setting of imperfect datasets.

5 Theoretically Principled Trade-off between Robustness and Accuracy

Challenge/Opportunity

- It remains unclear about how to trade off adversarial robustness against natural accuracy.
- The empiraical risk minimization (ERM) formulation of robust classification involves minimizing the robust 0-1 loss $\max_{\mathbf{x}':\|\mathbf{x}'-\mathbf{x}\|\leq\epsilon} \mathbf{1}\{c(\mathbf{x}') \neq y\}$ which is NP-hard to optimize even if $\epsilon = 0$ in general.
- Few theoretical work have been done on the theoretical side of the accuracy-robustness tradeoff.

SOA Knowledge/Solution

- Methods such as **relaxation based defenses** provide provable guarantees for adversarial robustness but ignore the natural accuracy.
- Some prior work on adversarial defense replaced the 0-1 loss $\mathbf{1}(\cdot)$ with a surrogate loss with little theoretical guarantee on the tightness of the approximation.
- One of the best known algorithms for adversarial defense is based on **robust optimization** with an objective function possibly not tight or capturing the accuracy-robustness tradeoff.
- Another is adversarial training via a regularization term that measures either the difference between $f(\mathbf{X})$ and Y or its gradient.
- **Adversarial Logit Pairing (ALP)**: the algorithm repeatedly generates an adversarial example $\mathbf{X}'$ by solving $\arg\max_{\mathbf{X}'\in\mathbb{B}(\mathbf{X},\epsilon)} \phi(f(\mathbf{X}')Y)$ and updating the network by solving $\min_f \mathbb{E}\{\alpha\phi(f(\mathbf{X}')Y) + (1-\alpha)\phi(f(\mathbf{X})Y) + \|f(\mathbf{X}) - f(\mathbf{X}')\|_2/\lambda\}$, where $\alpha \in [0,1]$ is a regularization parameter, until convergence.

Approach

- Theory
  - Define robust error under the threat model of bounded ϵ perturbation:

    $$\mathcal{R}_{\text{rob}}(f) = \mathbb{E}_{(\mathbf{X},Y)\sim\mathcal{D}}\mathbf{1}\{\exists\mathbf{X}' \in \mathbb{B}(\mathbf{X},\epsilon) \text{ s. t. } f(\mathbf{X'})Y \leq 0\}$$

    where $\mathbb{B}(\mathbf{X},\epsilon)$ is the ϵ-neighborhood of $\mathbf{X}$. Natural error:

    $$\mathcal{R}_{\text{nat}}(f) = \mathbb{E}_{(\mathbf{X},Y)\sim\mathcal{D}}\mathbf{1}\{f(\mathbf{X})Y \leq 0\}$$

    Boundary error:

    $$\mathcal{R}_{bdy}(f) = \mathbb{E}_{(\mathbf{X},Y)\sim\mathcal{D}}\mathbf{1}\{\mathbf{X} \in \mathbb{B}(\text{DB}(f),\epsilon), f(\mathbf{X})Y \leq 0\}$$

    where $\text{DB}(f)$ is the decision boundary of f. The robust error can be decomposed as the sum of the natural error and boundary error

    $$\mathcal{R}_{\text{rob}}(f) = \mathcal{R}_{\text{nat}}(f) + \mathcal{R}_{\text{bdy}}(f)$$

  - Provide a differentiable upper bound using classification-calibrated loss

$$\mathcal{R}_{\text{rob}}(f) - \mathcal{R}^*_{\text{nat}} \leq \phi^{-1}(\mathcal{R}_\phi(f) - \mathcal{R}^*_\phi) + \mathbb{E} \max_{\mathbf{X}' \in \mathbb{B}(\mathbf{X}, \epsilon)} \phi(f(\mathbf{X}')f(\mathbf{X})/\lambda)$$

- Design a new defense method TRADES to trade adversarial robustness off against accuracy with objective

$$\min_f \mathbb{E} \left\{ \underbrace{\phi(f(\mathbf{X})Y)}_{\text{accuracy}} + \underbrace{\max_{\mathbf{X}' \in \mathbb{B}(\mathbf{X}, \epsilon)} \phi(f(\mathbf{X})f(\mathbf{X}')/\lambda)}_{\text{regularization for robustness}} \right\}$$

  - extend to multi-class problems by involving multi-class calibrated loss
  - approximately solving the minimax problem using projected gradient descent
  - perform numerical experiments on MNIST and CIFAR10
  - use FGSM^k white-box and black-box method to attack various defense models and compare their performance with TRADES

Results

- Characterize the trade-off between accuracy and robustness for classification problems by decomposing the robust error as the sum of natural error and boundary error and provide a tightest differentiable upper bound (uniform over all probability distributions and measurable predictors).
- TRADES outperforms SOA methods under both black-box and white-box threat models.
- As the regularization parameter $1/\lambda$ increases, the natural accruacy decreases and the robust accuracy increases.
- Experiments show that TRADES are more robust against black-box attacks and can generate stronger adversarial examples for black-box attacks.

Thoughts

- The solution is to modify the objective function for improving robustness against adversarial examples.

Week 2

1 Assessing the Robustness of Visual Question Answering

Challenge/Opportunity

- There is a lack of analysis in the attacks at the image and/or the query, dubbed main question in VQA.
- Since robustness analysis needs to study the accuracy of VQA models under different noise levels, the level of noise for the given question should be controllable and quantified.
- Existing sentence evaluation metrics cannot capture the semantic meaning of text well and are ineffective.

SOA Knowledge/Solution

- Existing sentence evaluation metrics:
    - BLEU (overlapping of two texts - text similarity at corpus level)
    - METEOR (harmonic mean or unigram precision and recall, can handle the stemming and synonym matching - text similarity at sentence and segment level)
    - ROUGE (recall-based, reward longer sentences with higher recall)
    - CDEr (consensus-based, reward sentence similar to descriptions written by human expert, suit for image captioning but inefficient for natural language sentence evaluation)
- Previous works exploit language model, CNN+LSTM, attention mechanism, and deep neural networks for image captioning task.
- Previous works on VQA focus on modeling the interactions between the embedding spaces of images and texts using CNN, RNN, and **Gated Recurrent Unit (GRU)**.
- Recent works analyze the robustness of models by observing the consequence of adding noise or perturbations into images and provide theoretical relation between small curvature and large robustness but they are all restricted to adding noise to the image input.
- Datasets for VQA:
    - DAQUAR (a question only has a single ground truth answer)

    - Three VQA datasets based on MS-COCO
    - VQA (real-world images and abstract scenes, 10 answers for each image)
    - Visual Madlibs (multiple choice question answering)
    - Visual7W (answers+grounding, extended from Visual Genome dataset)
    - Xplore-M-Ego (natural language queries, media retrieval system, and collective memories, answers conditioned on questions and geographical position of the questioner)
- Another similar task is video question answering that requires understanding of long-term relations in the video (infer the past, describe the present, predict the future).

Approach

- Assume smaller text similarity score means more noise to the main question (MQ)
- Propose a new text ranking method using LASSO optimization

$$\min_{\mathbf{x}} \frac{1}{2}\|\mathbf{A}\mathbf{x}-\mathbf{b}\|_2^2+\lambda\|\mathbf{x}\|_1$$

  where λ is a tradeoff parameter that controls the quality of basic questions (BQs) (chosen to be 10^{-6}).
- Propose two large-scale basic questions datasets GBQD and YNBQD
    - use Skip-thoughts with a focus on semantic meaning to encode questions
    - combine the unique uestions in the training and validation datasets of VQA
    - do "question sentences preprocessing" such that none of MQs is contained in the BQD because of the sparsity encouraged by LASSO
    - collect top-21 BQs of a given MQ based on teh ranking scores
- Propose a method to measure the robustness of VQA models that consists of
    - a VQA module that will be analyzed for robustness
    - a level-controllable Noise Generator that takes a plain text main question, selects the topic three basic questions by their similarity rankings to MQ using a text similarity ranking method, and outputs their concatenation

- A predicted answer is considered correct if ≥ 3 annotators agree with it, which gives a measure for accuracy

$$\frac{1}{N}\sum_{i=1}^{N}\min\left\{\frac{\sum_{t\in T_i}\mathbb{I}[a_i=t]}{3},1\right\}$$

The robustness score is sensitive if the difference of accuracy on clean and perturbed datasets is small (not before t) and less sensitive if it is large (not after m):

$$R_{score}=\max\left(0,\min\left(1,\left(\frac{\sqrt{m}-\sqrt{Acc_{di}}}{\sqrt{m}-\sqrt{t}}\right)\right)\right)$$

where $Acc_{di}=|Acc_{vqa}-Acc_{bqd}|$.

- Propose Visual Question Answering by Basic Questions (VQABQ) framework to analyze existing VQA models.

Results

- The rankings by LASSO ranking method are effective on both GBQD and YNBQD datasets, which gives monotone scores consistent with the assumption, while the other seven existing metrics are not.
- The attention-based VQA models are more robust than non-attention-based ones in general.
- The variety of BQ candidates affects the robustness of attention-based VQA models in some sense.
- HieCoAtt is the most robust and MUTAN with Attention is the most accurate VQA model.
- If BQs are good enough, then direct concatenation of MQ and BQs helps the accuracy of VQA model.
- λ is one of the most important factors affecting the quality of BQs.

Thoughts

- This work is similar to what I want to achieve in my current research but I need to extend the domain beyond VQA.
- The second last result claim is weakly supported by empirical evidence.
- The adversarial attacks come from the dataset.

Challenge/Opportunity

SOA Knowledge/Solution

Approach

Results

Thoughts